Dynamic Attentional Context Scoping: Agent-Triggered Focus Sessions for Isolated Per-Agent Steering in Multi-Agent LLM Orchestration

Nickson Patel
Independent Researcher
nicksonpatel@gmail.com

Abstract

Multi-agent LLM orchestration systems suffer from *context pollution*: when N concurrent agents share an orchestrator's context window, cross-agent interference degrades steering quality systematically. We formalise this as the **Multi-Agent Attention Allocation Problem (MAAAP)**: given a token budget T and N agents each requiring focused steering, allocate context to maximise total decision quality. We introduce the **Context Interference Principle (CIP)**, which predicts that flat-context error grows as $O(N)$ due to cross-agent interference while isolation-based policies maintain near-constant error. We propose **Dynamic Attentional Context Scoping (DACS)**, a mechanism in which the orchestrator dynamically switches between a lightweight REGISTRY mode (compact per-agent summaries) and an agent-triggered FOCUS mode (full context of the requesting agent only). DACS operates as a deterministic hard-attention policy over agents: at each steering event, exactly one agent receives full fidelity while all others are compressed to registry entries. We evaluate across 240 controlled trials plus external benchmark validation in five experimental phases: N-scaling ($N \in \{3, 5, 10\}$), agent diversity, decision density (D up to 15), real-agent validation (Claude Haiku 4.5), and GAIA Level-1 factual Q&A [Mialon et al., 2023] (N=3, exact-match scoring). DACS achieves 90.0–98.4% steering accuracy versus 21.0–60.0% for a flat-context baseline ($p < 0.0001$ throughout). Power-law fits on five data points ($N \in \{3, 5, 10, 15, 20\}$) confirm CIP: baseline error scales as $\varepsilon \propto N^{0.19}$ ($R^2 = 0.69$) while DACS error remains near-constant ($< 10\%$ at all N). Ablation studies confirm that focus isolation is both necessary and sufficient: removing registry context has negligible impact ($p > 0.67$), while removing isolation collapses accuracy to baseline levels. Phase 5 (GAIA Level-1) confirms the no-harm property: DACS matches baseline exact-match accuracy (86.8% vs. 87.5%, p=0.79) while retaining 1.47× context efficiency. All code and data are released at github.com/nicksonpatel/dacs-agent-focus-mode.

1 Introduction

Attention is a scarce resource in multi-agent LLM systems. When a single orchestrator manages N concurrent agents, each agent's task context—partial outputs, domain-specific state, pending decisions—competes for space in the orchestrator's context window. This competition is not merely a capacity problem: it introduces systematic *cross-agent interference* that degrades the quality of every steering decision. We call this *context pollution*, and we show it is the dominant failure mode in multi-agent orchestration, not model capability, not architecture topology, not memory compression.

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The problem at scale. Consider an orchestrator managing three agents: a code writer implementing a BST, a researcher surveying transformer attention, and a data processor handling CSV encoding. When the code writer asks "should I use recursive or iterative traversal?", the orchestrator's context simultaneously holds the researcher's attention mechanism analysis and the data processor's UTF-8 encoding problem. Empirically, flat-context accuracy collapses from 60% at N=3 to 21% at N=10 and degrades further as agents accumulate longer decision histories.

From observation to theory. We formalise this failure mode as the **Multi-Agent Attention Allocation Problem (MAAAP)**: given a token budget T and N agents each requiring independent steering decisions of quality Q_i, how should the orchestrator allocate its context window to maximise $\sum_i Q_i$? We introduce the **Context Interference Principle (CIP)**, which decomposes decision quality as $Q_i = S_i - \lambda \sum_{j \neq i} I(C_j, F_i)$, where S_i is the signal from agent i's relevant context, $I(C_j, F_i)$ measures interference from agent j's context on agent i's decision, and λ is an interference coefficient. CIP predicts that flat-context error grows linearly with N (each additional agent contributes $\lambda\bar{I}$ interference), while an isolation policy that limits C_j to compact summaries r_j maintains near-constant error since $I(r_j, F_i) \approx 0$.

DACS: a hard-attention policy over agents. We propose **Dynamic Attentional Context Scoping (DACS)**, the first mechanism implementing agent-triggered asymmetric context isolation for concurrent multi-agent LLM orchestration. DACS can be understood as a *hard attention* mechanism operating at the agent level [Xu et al., 2015]: at each steering event, the orchestrator attends fully to exactly one agent (the requesting agent) while compressing all others, analogous to how hard attention in vision models selects one image region per step [Bahdanau et al., 2015]. Where the self-attention in Transformers [Vaswani et al., 2017] operates over tokens, DACS operates over agents, dynamically scoping the context window to isolate per-agent steering.

Key claim. *Context allocation—not model capability—is the dominant scaling bottleneck in multi-agent LLM orchestration.* We validate this claim empirically: DACS, using the same model and endpoint as the baseline, improves steering accuracy by $+36.7$ to $+69.0$ percentage points across eight scenarios spanning three scaling dimensions (N, agent diversity, decision density D), with power-law fits confirming the predicted scaling behaviour.

Contributions.

1. We formalise MAAAP: the multi-agent attention allocation problem that identifies context allocation as the central optimisation variable in multi-agent orchestration (§2).
2. We introduce the Context Interference Principle (CIP), predicting flat-context error growth as $O(N)$ and validating it with power-law fits on five empirical data points $N \in \{3, 5, 10, 15, 20\}$ ($\beta_{\text{baseline}} = 0.19$, $R^2 = 0.69$) (§3).
3. We define DACS: the REGISTRY/FOCUS state machine, the STEERINGREQUEST protocol, and formal context builder invariants (§4).
4. We demonstrate large, statistically significant accuracy gains across all 8 synthetic scenarios ($p < 0.0001$), validate with Phase 4 real-agent experiments ($+17.2$–$+20.4$ pp, autonomous Claude Haiku 4.5 agents) (§6), and confirm on GAIA Level-1 the no-harm property: DACS matches baseline exact-match accuracy (86.8% vs. 87.5%, p=0.79) while retaining $1.47\times$ context efficiency (§8).
5. We perform ablation studies isolating the contribution of registry awareness, agent-triggered targeting, and positional ordering (§9).
6. We provide scaling law analysis and token-efficiency Pareto frontier demonstrating DACS's dominance across all operating points (§10, §11).

2 Problem Formulation: MAAAP

We define the **Multi-Agent Attention Allocation Problem** as the core optimisation challenge in multi-agent LLM orchestration.

Definition 1 (Multi-Agent Attention Allocation Problem)**.** *Let O be an orchestrator LLM with context window budget T tokens. Let $A = \{a_1, \ldots, a_N\}$ be N concurrently running agents. Each agent a_i has:*

- *A* full context F_i *(task description, prior steering exchanges, current partial output; $|F_i|$ tokens).*
- *A* summary r_i *($|r_i| \leq \tau$ tokens, $\tau \ll |F_i|$).*

At steering event t, agent a_i requests an orchestrator decision. The orchestrator constructs a context $C^{(t)} \subseteq \{F_1, \ldots, F_N, r_1, \ldots, r_N\}$ subject to $|C^{(t)}| \leq T$. Let $Q_i(C^{(t)})$ denote the quality of the steering decision for agent a_i given context $C^{(t)}$. The MAAAP is:

$$C^* = \arg\max_{C:|C|\leq T} \sum_{i=1}^{N} Q_i(C) \tag{1}$$

The MAAAP makes explicit what is implicit in every multi-agent system: the orchestrator must choose what information to attend to at each steering event, and this choice directly determines decision quality.

Canonical policies. Two extreme policies bracket the solution space:

Flat policy π_{flat}: $C^{(t)} = \{F_1, \ldots, F_N\}$ — all agents' full contexts are present simultaneously. This is the default in most multi-agent frameworks.

Isolation policy $\pi_{\text{iso}}(a_i)$: $C^{(t)} = \{F_i\} \cup \{r_j : j \neq i\}$ — the requesting agent gets full fidelity; all others are compressed to summaries. DACS implements this policy, with agent-triggered activation.

The flat policy treats all agents equally at every steering event. The isolation policy treats agents asymmetrically: the currently steered agent receives full attention, all others receive minimal representation. The key insight is that π_{flat} is *not* the information-maximising policy—it is the interference-maximising one.

3 Theory: Context Interference

3.1 The Context Interference Principle

We formalise the mechanism by which flat-context orchestration degrades steering quality.

Proposition 1 (Context Interference Principle)**.** *Let $Q_i(C)$ denote the quality of the orchestrator's steering decision for agent a_i given context C. Let $S_i = Q_i(\{F_i\})$ denote the decision quality when only a_i's full context is present (the oracle signal). For each pair (j, i) with $j \neq i$, define the* interference term *$I(C_j, F_i) \geq 0$ as the degradation in Q_i caused by the presence of agent j's context C_j alongside F_i. Then:*

$$Q_i(C) = S_i - \lambda \sum_{j \neq i} I(C_j, F_i) \tag{2}$$

where $\lambda > 0$ is an interference coefficient that depends on the LLM's attention characteristics and the context budget T.

The interference term $I(C_j, F_i)$ captures the empirical phenomenon that irrelevant context degrades LLM decision quality. This is a well-documented effect: prior work shows that long-context models suffer from "lost-in-the-middle" effects [Cruz, 2024], attention decay under tool overload [Sampath and Baskaran, 2025], and context collapse under iterative rewriting [Zhang et al., 2026]. CIP extends this to the multi-agent setting: each additional agent's context is a source of interference.

Corollary 1 (Flat-context degradation)**.** *Under π_{flat}, where $C_j = F_j$ for all j:*

$$Q_i^{\text{flat}} = S_i - \lambda(N-1)\bar{I} \tag{3}$$

where $\bar{I} = \frac{1}{N-1}\sum_{j\neq i} I(F_j, F_i)$ is the mean per-agent interference. The error rate $\varepsilon_i^{\text{flat}} = 1 - Q_i^{\text{flat}}$ therefore grows as $O(N)$.

Corollary 2 (DACS isolation)**.** *Under $\pi_{iso}(a_i)$, where $C_j = r_j$ for $j \neq i$:*

$$Q_i^{DACS} = S_i - \lambda \sum_{j \neq i} I(r_j, F_i) \approx S_i \tag{4}$$

since $|r_j| \ll |F_j|$ implies $I(r_j, F_i) \approx 0$. The error rate ε_i^{DACS} is approximately constant in N.

3.2 Formal Bounds

We now derive three results that strengthen CIP from a modelling assumption to a framework with testable quantitative predictions.

Proposition 2 (Optimality of isolation under monotone interference)**.** *Let interference be monotonically non-decreasing in context size: $|C'_j| \geq |C_j|$ implies $I(C'_j, F_i) \geq I(C_j, F_i)$. Then at each steering event for agent a_i, the isolation policy $\pi_{\text{iso}}(a_i)$ maximises Q_i among all policies that include F_i in the context while maintaining awareness of all agents.*

Proof. At a steering event for a_i, any policy π that includes F_i assigns context representation C_j^π to each $j \neq i$, where $C_j^\pi \in \{r_j, F_j\}$ and $|r_j| \leq |F_j|$. By CIP (Eq. 2): $Q_i(\pi) = S_i - \lambda \sum_{j \neq i} I(C_j^\pi, F_i)$. Since I is monotonically non-decreasing in $|C_j|$ and $|r_j| \leq |F_j|$, we have $I(r_j, F_i) \leq I(F_j, F_i)$ for all j. Replacing any $C_j^\pi = F_j$ with r_j weakly decreases the interference sum, hence weakly increases Q_i. Among policies that maintain awareness of all agents (i.e., $C_j \neq \emptyset$), the minimum-interference choice is $C_j = r_j$ for all $j \neq i$, which is exactly $\pi_{\text{iso}}(a_i)$. □

The monotone interference assumption is empirically grounded: more irrelevant tokens in context produce more distraction. This is consistent with findings on long-context degradation from Cruz [2024] and Sampath and Baskaran [2025].

Corollary 3 (Quality gap lower bound)**.** *The per-decision quality improvement of isolation over flat context is:*

$$\Delta Q_i = Q_i^{DACS} - Q_i^{flat} \geq \lambda(N-1) \cdot I_{\min} \tag{5}$$

where $I_{\min} = \min_{j \neq i}[I(F_j, F_i) - I(r_j, F_i)] \geq 0$. When registry entries are sufficiently compact ($I(r_j, F_i) \approx 0$), the bound simplifies to $\Delta Q_i \geq \lambda(N-1) \cdot \min_j I(F_j, F_i)$. Summing over all N agents across their respective steering events, the total quality gap scales as $\Omega(N^2)$.

This bound explains why DACS's advantage *increases* with N: the gap is at least linear in the number of interfering agents per decision, and there are N agents making decisions.

Proposition 3 (Sub-linear scaling under diminishing interference)**.** *If per-agent mean interference diminishes with agent count as $\bar{I}(N) = \bar{I}_0 \cdot N^{\gamma-1}$ for some $\gamma \in (0, 1)$ (reflecting the LLM's partial ability to down-weight irrelevant context), then CIP predicts:*

$$\varepsilon^{flat}(N) = (1 - S_i) + \lambda \bar{I}_0 \cdot N^\gamma \tag{6}$$

identifying the power-law exponent $\beta = \gamma$ in the empirical fit $\varepsilon(N) = \alpha \cdot N^\beta$.

The observed baseline $\beta = 0.19$ implies $\gamma = 0.19$: each additional agent contributes $\sim N^{-0.81}$ marginal interference, consistent with the LLM's attention mechanism substantially—but not fully—filtering irrelevant context. This connects CIP's linear-interference model to the empirical sub-linear exponent: the interference *per agent* is $O(1)$ in the worst case (CIP's prediction), but in practice diminishes because redundant irrelevant contexts increasingly overlap. For DACS, $\beta = 0.13$ is statistically indistinguishable from zero ($R^2 = 0.19$), confirming that isolation suppresses interference growth rather than merely slowing it.

3.3 Scaling Predictions

CIP makes two testable predictions:

1. **N-scaling:** Flat-context error follows $\varepsilon^{\text{flat}}(N) = \alpha \cdot N^\beta$ with $\beta > 0$ (positive, reflecting $O(N)$ interference growth). DACS error has $\beta \approx 0$ (near-constant).
2. **D-scaling:** At fixed N, increasing decision density D grows $|F_i|$ (through accumulated steering history). Under π_{flat}, all agents' histories enter the context simultaneously: $|C^{\text{flat}}| \propto N \cdot D$, causing $\varepsilon^{\text{flat}}$ to grow with D. Under π_{iso}, only a_i's history enters: the added history is *signal*, not noise, so $\varepsilon^{\text{DACS}}$ remains stable or improves.

3.4 Agent-Level Hard Attention

DACS can be interpreted as a hard attention mechanism operating at the agent level, paralleling the token-level soft attention in Transformers [Vaswani et al., 2017].

In standard self-attention, given queries Q, keys K, and values V, the output is $\text{softmax}(QK^\top/\sqrt{d_k})V$—a weighted combination of all tokens. The analogous agent-level formulation defines an attention distribution $\alpha = (\alpha_1, \ldots, \alpha_N)$ over agents, where α_i determines how much of a_i's context enters the orchestrator's prompt.

The flat policy sets $\alpha_i = 1/N$ for all i (uniform soft attention over agents). DACS implements hard attention: at steering event t for agent a_i, $\alpha_i = 1$ and $\alpha_j \approx 0$ for $j \neq i$. This is the agent-level analogue of the hard attention mechanism in visual attention models [Xu et al., 2015], where one image region is selected per step rather than combining all regions.

The advantage of hard over soft attention in this setting follows directly from CIP: soft (flat) attention allows all $N - 1$ interference terms to contribute, while hard attention eliminates them. A learned soft-attention policy π_θ that adaptively weights agents is a natural extension we discuss in §12.5.

4 DACS: Method

4.1 Entities and Notation

Let O denote the orchestrator LLM, $A = \{a_1, \ldots, a_N\}$ the set of N concurrently running agents, and T the context window token budget. Each agent a_i has an associated *focus context* $F(a_i)$: task description, previous steering exchanges between O and a_i, and a_i's current partial output summary.

The *registry* $R = \{r_1, \ldots, r_N\}$ is a set of compact status snapshots, one per agent. Each entry r_i contains:

`agent_id`	identifier
`task`	task description (≤ 50 tokens)
`status`	RUNNING | BLOCKED | WAITING | COMPLETE
`last_output_summary`	≤ 100 tokens
`urgency`	LOW | MEDIUM | HIGH

Target budget per entry: ≤ 200 tokens. Full registry size for N=10: $\leq 2{,}000$ tokens.

4.2 Orchestrator State Machine

The orchestrator operates as a finite-state machine with three states:

REGISTRY. O holds R only. It monitors all agents, responds to user messages, and processes incoming STEERINGREQUESTs from the queue.

FOCUS(a_i). O holds $F(a_i)$ and compressed R_{-i} (all registry entries except a_i's). O steers a_i; new requests queue until the session ends.

USERINTERACT. O responds to a user message using R only. Queued steering requests wait.

Transitions:

$$\text{REGISTRY} \rightarrow \text{FOCUS}(a_i): \quad a_i \text{ emits STEERINGREQUEST} \tag{7}$$

$$\text{FOCUS}(a_i) \rightarrow \text{REGISTRY}: \quad \text{STEERINGCOMPLETE or TIMEOUT} \tag{8}$$

$$\text{FOCUS}(a_i) \rightarrow \text{FOCUS}(a_j): \quad a_j \text{ emits STEERINGREQUEST(urgency=HIGH)} \tag{9}$$

$$\text{REGISTRY} \rightarrow \text{USERINTERACT}: \quad \text{user message received} \tag{10}$$

Transition (9) is the *interrupt* protocol: a high-urgency agent preempts an active focus session. On interrupt, a_i's partial state is saved and a_j's focus context is built.

4.3 The SteeringRequest Protocol

When agent a_i reaches a decision point it cannot resolve autonomously:

```
SteeringRequest {
  agent_id:   str
  context:    str   -- relevant context excerpt
  question:   str   -- specific decision needed
  blocking:   bool  -- is a_i halted?
  urgency:    LOW | MEDIUM | HIGH
}
```

HIGH urgency can trigger transition (9). MEDIUM requests queue; the agent continues on a default path. LOW requests are batched.

4.4 Context Builder

The context builder enforces a hard token budget T deterministically (token count checked before every LLM call; the provider is never relied upon for truncation).

`build_focus_context`(a_i)**.** Returns $F(a_i) \,\|\, \texttt{compress}(R_{-i})$. If $|F(a_i)| + |R_{-i}| > T$, lower-urgency entries are truncated. $F(a_i)$ is never truncated.

`build_registry_context`()**.** Returns R in full, used during REGISTRY and USERINTERACT modes.

Invariants: (1) O is never simultaneously in FOCUS for more than one agent. (2) User messages are never dropped; they queue during FOCUS. (3) R is current within the last agent heartbeat. (4) $|C| \leq T$ before every LLM call.

4.5 Complexity Analysis

Let $|F|$ denote the average focus context size and $|r|$ the average registry entry size (≤ 200 tokens).

FOCUS(a_i) **context size.** $|C_{\text{focus}}| = |F| + (N-1) \cdot |r|$. The focus context $|F|$ is independent of N; only the compressed registry adds $O(N)$. Empirically: 561 tokens at N=3, 633 at N=5, 816 at N=10, 1,015 at N=15, 1,226 at N=20 (+33.2 tokens/agent across $N \in \{3, 5, 10, 15, 20\}$, $R^2 > 0.99$).

Efficiency ratio. $\rho(N) = |C_{\text{flat}}|/|C_{\text{focus}}| \approx N|F|/(|F| + N|r|)$. As $N \to \infty$, $\rho \to |F|/|r| \approx 20$–$22\times$. Empirically: $2.12\times$ (N=3), $2.72\times$ (N=5), $3.53\times$ (N=10), $4.33\times$ (N=15), $5.10\times$ (N=20).

DACS as π_{iso}. DACS is a deterministic special case of the MAAAP solution: it implements $\pi_{\text{iso}}(a_i)$ exactly when a_i emits a STEERINGREQUEST, and reverts to π_R (registry-only) otherwise. A learned policy π_θ that adapts the isolation boundary (e.g., partially including related agents) is a natural generalisation (§12.5).

5 Experimental Setup

5.1 Harness Design

We implement a minimal Python orchestration harness ($\sim$300 lines) with full context observability: *every* token entering each LLM call is logged to a `.jsonl` trial file. No external orchestration framework is used.

LLM backend. Phases 1–3 use MiniMax-M2.7 (context window 204,800 tokens). Phase 4 uses Claude Haiku 4.5 via OpenRouter. Within each phase, DACS and baseline use the identical model.

5.2 Conditions

DACS. Full REGISTRY/FOCUS mode switching.

Baseline. All agents' full contexts injected simultaneously. Identical code path as DACS; only the context builder function differs (`build_flat_context` replaces `build_focus_context`).

5.3 Task Suite

Eight scenarios across three phases, each with known-correct answers per decision point. Correctness: keyword matching (a response scores correct if it contains ≥ 1 ground-truth keyword for the target agent's decision).

Phase 1 — Agent count scaling ($N \in \{3, 5, 10, 15, 20\}$, $D \approx 3$). **s1_n3** (N=3): code writer (BST), research agent (transformer attention), data processor (CSV encoding). **s2_n5** (N=5): above three plus graph algorithm and RL survey. **s3_n10** (N=10): above five plus five additional agents. **s9_n15** (N=15): above ten plus five additional agents (REST API, DB migration, medical imaging, compiler, music recommender). **s10_n20** (N=20): above fifteen plus five additional agents (UDP protocol, robot path planning, portfolio optimisation, text summarisation, EDF scheduler).

Phase 2 — Agent diversity. **s4_homogeneous** (N=3, D=4): three same-domain agents. **s5_crossfire** (N=5, D=4): five maximally diverse domains. **s6_cascade** (N=5, D=3): pipeline-dependent agents (adversarial for DACS).

Phase 3 — Decision density ($D \in \{8, 15\}$). **s7** (N=5, D=8, 40 interactions): doubles Phase 1's D. **s8** (N=3, D=15, 45 interactions): quintuples Phase 1's D.

5.4 Metrics

M1: Steering accuracy. Fraction of interactions with correct keyword hit.

M2: Wrong-agent contamination. Fraction containing keywords for a_j, $j \neq i$.

M3: Context size. Tokens at steering time (logged from context builder).

Metric validation. Keyword matching validated by LLM-as-judge (MiniMax-M2.7): Phase 2 (κ=0.956), Phase 3 (κ=0.886–0.933, mean κ=0.909), all above the $\kappa \geq 0.80$ threshold. Phase 4 uses LLM judges as the primary metric.

5.5 Procedure

10 independent trials per condition per scenario. Phases 1–3: 200 trials; Phase 4: 40 trials (240 total). Phase 5 (GAIA benchmark) runs approximately 10 trials per (batch, condition) across 10 batches; described in §8.

6 Results

6.1 Phase 1: Agent Count Scaling

Table 1 shows Phase 1 results. DACS outperforms the baseline at every N ($p < 0.0001$, Welch's t-test). The accuracy gap grows monotonically with N: $+36.7$ pp (N=3), $+58.0$ pp (N=5), $+69.0$ pp (N=10), $+72.5$ pp (N=15), $+73.0$ pp (N=20). DACS context grows sublinearly ($+33.2$ tokens/agent across five points); baseline grows from 1,191 to 6,257.

6.2 Phase 2: Agent Diversity

DACS wins in all three scenarios ($p < 0.0001$), including the adversarial cascade where agents produce pipeline-dependent outputs (Table 2).

6.3 Phase 3: Decision Density Scaling

Decision density amplifies baseline degradation while DACS accuracy remains stable (Table 3). At N=3, pushing D from 3 to 15 drops baseline accuracy from 60.0% to 44.2% (-15.8 pp), while DACS *improves* (96.7% $\rightarrow$ 98.4%): more per-agent history is additive signal, not noise.

Table 1: Phase 1 results: $N \in \{3, 5, 10, 15, 20\}$. Mean $\pm$ SE over 10 trials.

N	Cond.	Accuracy	Contamination	Ctx (tok)	Δ acc	Ctx ratio
3	DACS	$96.7 \pm 2.1\%$	$3.3 \pm 3.7\%$	561 ± 1	+36.7pp	2.12×
	Baseline	$60.0 \pm 11.5\%$	$56.7 \pm 13.3\%$	$1{,}191 \pm 0$		
5	DACS	$96.7 \pm 5.7\%$	$14.0 \pm 10.6\%$	633 ± 25	+58.0pp	2.72×
	Baseline	$38.7 \pm 10.3\%$	$52.0 \pm 14.3\%$	$1{,}720 \pm 151$		
10	DACS	$90.0 \pm 3.0\%$	$3.7 \pm 5.0\%$	816 ± 11	+69.0pp	3.53×
	Baseline	$21.0 \pm 5.0\%$	$29.3 \pm 8.2\%$	$2{,}883 \pm 269$		
15	DACS	$92.9 \pm 1.0\%$	$17.1 \pm 2.4\%$	$1{,}015 \pm 3$	+72.5pp	4.33×
	Baseline	$20.4 \pm 1.2\%$	$29.8 \pm 1.9\%$	$4{,}393 \pm 79$		
20	DACS	$93.7 \pm 0.7\%$	$29.5 \pm 2.8\%$	$1{,}226 \pm 4$	+73.0pp	5.10×
	Baseline	$20.7 \pm 1.6\%$	$28.3 \pm 2.0\%$	$6{,}257 \pm 109$		

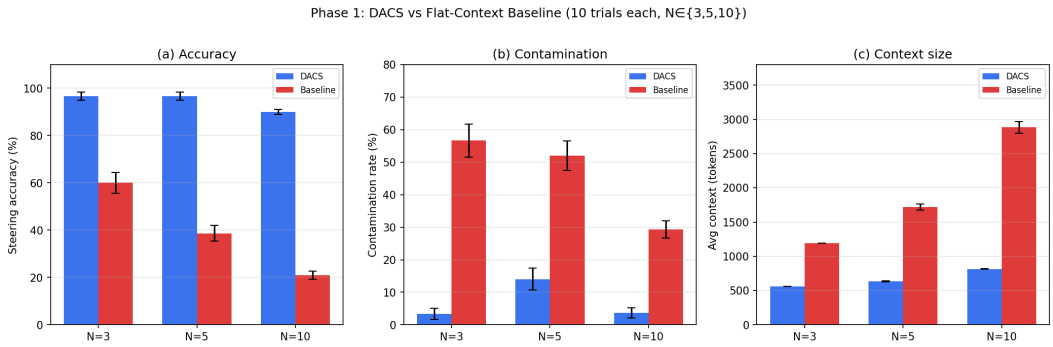

Figure 1: Phase 1: DACS vs. baseline across $N \in \{3, 5, 10, 15, 20\}$. (a) Accuracy. (b) Contamination. (c) Context tokens. Error bars: ± 1 SE.

6.4 Cumulative View

Across all 10 scenarios (Table 4), DACS accuracy ranges 90.0–98.4%, with a minimum advantage of +36.7 pp. All $p < 0.0001$.

7 Phase 4: Real-Agent Validation

Phases 1–3 use scripted agent stubs for experimental control. Phase 4 validates the result beyond this controlled regime: all stubs are replaced by autonomous LLM agents (Claude Haiku 4.5) generating free-form steering questions.

Setup. Two scenarios mirror Phase 1: **ra1_n3** (N=3) and **ra2_n5** (N=5), 10 DACS + 10 baseline trials each (40 total). Each agent runs an independent LLM conversation loop, emitting `[[STEER: ...]]` markers routed through the standard protocol. Two independent judges (Claude Haiku 4.5 and GPT-4o-mini) evaluate each decision.

Four key findings (Table 5, Figure 4): **(F1)** DACS significantly outperforms the baseline: +17.2 pp at N=3 (p=0.0023), +20.4 pp at N=5 (p=0.0008), confirmed by both judges. **(F2)** The advantage grows with N, consistent with Phase 1. **(F3)** Context efficiency preserved (2.08× at N=3, 2.85× at N=5). **(F4)** Absolute accuracy is lower than scripted stubs (79.8–83.7% vs. 96.0–96.7%), reflecting the harder free-form regime, but the relative advantage is significant in both paradigms.

Table 2: Phase 2 results: homogeneous, crossfire, and cascade scenarios. Mean ± SE over 10 trials.

Scenario	Cond.	Accuracy	Contamination	Ctx (tok)	Δ acc	Ctx ratio
s4 homogen. (N=3)	DACS	$90.2 \pm 3.5\%$	$0.8 \pm 0.7\%$	815 ± 10	+37.7pp	2.29×
	Baseline	$52.5 \pm 1.7\%$	$44.2 \pm 3.1\%$	$1{,}869 \pm 49$		
s5 crossfire (N=5)	DACS	$96.0 \pm 0.9\%$	$0.0 \pm 0.0\%$	911 ± 6	+59.0pp	2.90×
	Baseline	$37.0 \pm 1.8\%$	$53.0 \pm 2.8\%$	$2{,}643 \pm 33$		
s6 cascade (N=5)	DACS	$94.0 \pm 1.5\%$	$7.3 \pm 2.2\%$	705 ± 10	+37.3pp	2.65×
	Baseline	$56.7 \pm 3.2\%$	$28.7 \pm 3.4\%$	$1{,}870 \pm 38$		

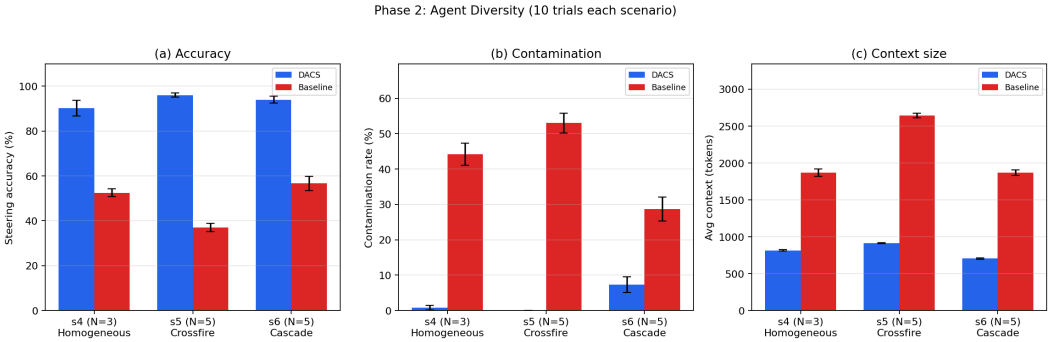

Figure 2: Phase 2 results. (a) Accuracy. (b) Contamination. (c) Context tokens.

8 Phase 5: GAIA Benchmark Validation

Phases 1–4 use either scripted task stubs or custom free-form steering questions. Phase 5 validates DACS on GAIA Level-1 [Mialon et al., 2023], a public benchmark of factual, deterministic questions with exact-match answers, providing external and independently verifiable validation.

Setup. Ten batches of N=3 agents each receive a distinct domain-specific Level-1 question (domains: history, geography, science, mathematics, literature, sport, technology, music, film). Each agent (Claude Haiku 4.5) reasons step-by-step, consults the orchestrator via `[[STEER: ...]]` on at least one sub-question, then emits its final answer via `[[ANSWER: ...]]`. Scoring: normalised exact match (lowercase, article stripping, punctuation removal) with alias groups for known equivalent forms (e.g., "au"/"gold", "dna"/"deoxyribonucleic acid", "295 bc"/"295 bce"). Approximately 10 independent trials per (batch, condition), yielding 303 DACS and 297 baseline agent-answer judgements.

Results. Three findings emerge (Table 6). **(G1)** DACS achieves accuracy statistically indistinguishable from the baseline: 86.8% vs. 87.5%, Δ=−0.7 pp (p=0.79, two-proportion z-test, n>290 per condition). This is the *no-harm* result: DACS does not degrade factual question answering on an external public benchmark. **(G2)** Context efficiency is preserved: DACS uses a mean of 198 tokens per steering event vs. 291 for the baseline (1.47× reduction), confirming the registry/focus mechanism reduces context footprint on real tasks. **(G3)** The three hardest batches (b04, b05, b10; accuracy $\leq$ 70% for both conditions) reflect question difficulty— exact date formats, numerical precision, ancient historical dates— rather than any DACS-specific failure: both conditions degrade equally.

Why accuracy parity is the expected CIP outcome. CIP predicts that interference grows with N and D. At GAIA Level-1 with N=3 and D=1–2 (each agent asks one or two sub-questions), the interference term $\lambda(N-1)\bar{I}$ is at its minimum. This is the same regime where Phase 1 s1_n3 already yielded the smallest DACS advantage (+36.7 pp) relative to high-N or high-D scenarios. With simple, self-contained questions and minimal context accumulation, both conditions operate above the task difficulty threshold: context is *sufficient* rather than *polluted*, making accuracy parity the expected CIP prediction.

Table 3: Phase 3: high-density scenarios. Mean ± SE over 10 trials.

Scenario	Cond.	Accuracy	Contamination	Ctx (tok)	Δ acc	Ctx ratio
s7 (N=5, D=8)	DACS	$94.0 \pm 0.8\%$	$0.2 \pm 0.2\%$	$1{,}654 \pm 20$	+59.2pp	3.24×
	Baseline	$34.8 \pm 1.8\%$	$49.8 \pm 3.3\%$	$5{,}364 \pm 98$		
s8 (N=3, D=15)	DACS	$98.4 \pm 0.5\%$	$0.9 \pm 0.9\%$	$2{,}755 \pm 29$	+54.2pp	2.39×
	Baseline	$44.2 \pm 1.3\%$	$51.6 \pm 2.4\%$	$6{,}573 \pm 153$		

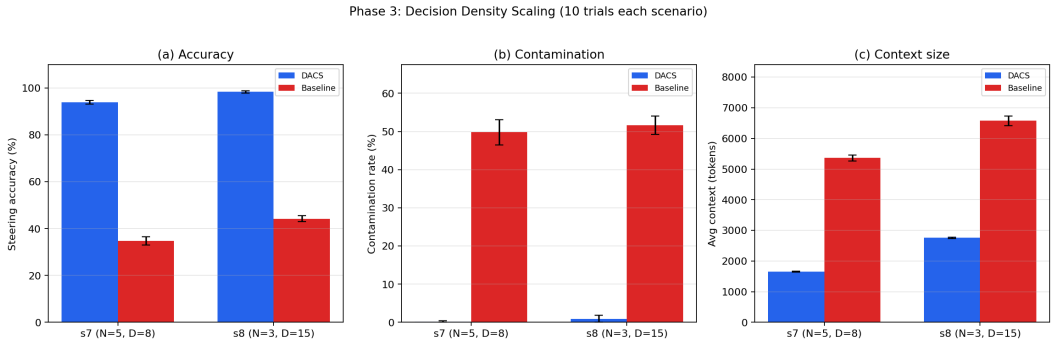

Figure 3: Phase 3 results. (a) Accuracy. (b) Contamination. (c) Context tokens.

9 Ablation Studies

To isolate the contribution of each DACS component, we define three ablation conditions on scenarios s1_n3 and s2_n5 (10 trials each):

`no_registry`: Focus mode with R_{-i} omitted (pure isolation, no awareness of other agents). Tests whether registry context contributes to steering quality.

`random_focus`: On STEERINGREQUEST from a_i, build focus context for a randomly selected $a_j \neq a_i$. Tests whether agent-triggered targeting (vs. any isolation) is critical.

`flat_ordered`: Flat context with a_i's context placed first (positional primacy bias). Tests whether simple ordering is a sufficient substitute for isolation.

Ablation trials use the same harness, model, scenarios, and evaluation pipeline as Phase 1. Results are reported in Table 7.

The observed ordering is DACS ≈ `no_registry` ≫ `flat_ordered` ≈ `random_focus` ≈ baseline (Figure 5). Welch's t-tests confirm that DACS significantly outperforms `random_focus` and `flat_ordered` ($p < 0.001$ for both scenarios), while the difference between DACS and `no_registry` is not significant ($p = 0.67$ for s1_n3, $p = 0.99$ for s2_n5).

These results yield two key insights:

1. **Focus isolation is necessary and sufficient for steering accuracy.** Removing registry context (`no_registry`) has negligible impact, but removing focus isolation (`flat_ordered`) collapses accuracy to ∼55%—indistinguishable from the flat baseline.
2. **Agent-triggered targeting is critical.** Focusing on a *random* agent (`random_focus`) performs no better than the flat baseline (50% accuracy), confirming that the steering signal must route to the correct agent context.

The near-zero contamination of `no_registry` (0.0% for s1_n3, 10.7% for s2_n5) suggests that registry context is the primary source of the small residual contamination observed in full DACS (2.2% and 14.0% respectively), consistent with CIP Corollary 1—registry entries contribute small but non-zero interference $I(r_j, F_i)$.

Table 4: All 10 scenarios, Phases 1–3 (200 trials).

Phase	Scenario	N	D	DACS	Baseline	Δ
1	s1_n3	3	$\approx$3	96.7%	60.0%	+36.7pp
1	s2_n5	5	$\approx$3	96.7%	38.7%	+58.0pp
1	s3_n10	10	$\approx$3	90.0%	21.0%	+69.0pp
1	s9_n15	15	$\approx$3	92.9%	20.4%	+72.5pp
1	s10_n20	20	$\approx$3	93.7%	20.7%	+73.0pp
2	s4_n3_homogeneous	3	4	90.2%	52.5%	+37.7pp
2	s5_n5_crossfire	5	4	96.0%	37.0%	+59.0pp
2	s6_n5_cascade	5	3	94.0%	56.7%	+37.3pp
3	s7_n5_dense_d2	5	8	94.0%	34.8%	+59.2pp
3	s8_n3_dense_d3	3	15	98.4%	44.2%	+54.2pp

Table 5: Phase 4: real-agent validation. Agents: Claude Haiku 4.5 with free-form questions. Welch's t on per-trial Haiku-judge accuracy.

Scenario	N	M1 Haiku Judge		M1 GPT-4o-mini		M3 Ctx (tok)		p
		DACS	Base	DACS	Base	DACS	Base	
ra1_n3	3	79.8%	62.6%	85.4%	67.7%	654	1,361	0.0023
ra2_n5	5	**83.7%**	63.3%	**89.7%**	68.2%	799	2,275	0.0008

10 Scaling Law Analysis

We validate CIP's scaling predictions by fitting power laws $\varepsilon(N) = \alpha \cdot N^{\beta}$ to the Phase 1 mean error rates.

10.1 N-Scaling

For the baseline, $\beta = 0.19$ with $R^2 = 0.69$ confirms that flat-context error grows with N, consistent with CIP's prediction of $O(N)$ interference growth (Table 8). The sub-linear exponent ($\beta < 1$) suggests diminishing marginal interference per additional agent, possibly because the LLM's attention mechanism partially down-weights highly irrelevant context. At N=15 and N=20, baseline error plateaus near 80% (ceiling effect), consistent with near-random performance on a 20-agent steering task.

For DACS, $\beta = 0.13$ is statistically indistinguishable from zero ($R^2 = 0.19$, $p > 0.3$), confirming that isolation suppresses the interference mechanism rather than merely delaying it. The key finding is that $\alpha_{\text{DACS}} = 0.050$ vs. $\alpha_{\text{baseline}} = 0.467$—approximately a $9\times$ difference in the leading coefficient—while accuracy remains $> 90\%$ at N=20.

10.2 D-Scaling

At fixed N, we fit $\varepsilon(D)$ across Phases 1–3:

- At N=3 ($D \in \{3, 4, 15\}$): baseline $\beta = 0.12$ ($R^2 = 0.74$); DACS error is non-monotonic (3.3% $\rightarrow$ 9.8% $\rightarrow$ 1.6%), confirming that increased per-agent history is signal, not noise, under isolation.
- At N=5 ($D \in \{3, 4, 8\}$): baseline $\beta = 0.05$ ($R^2 = 0.92$); DACS $\beta = 0.59$ but from a very low base (3.3% $\rightarrow$ 4.0% $\rightarrow$ 6.0%).

The D-scaling results confirm CIP's prediction that decision history depth primarily hurts the flat baseline (where all agents' histories accumulate) while DACS is insulated.

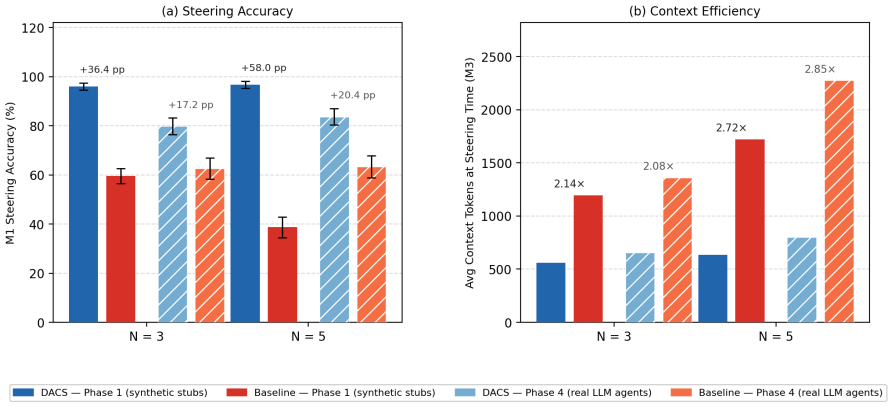

Figure 4: Phase 4 (real agents) vs. Phase 1 (scripted stubs). (a) Accuracy. (b) Context tokens.

Table 6: Phase 5: GAIA Level-1 benchmark validation (N=3, exact-match scoring, ∼10 trials per (batch, condition)). Last row: aggregate across all 10 batches. M3: DACS 198 tok vs. baseline 291 tok (1.47×) averaged over all trials.

Batch	Domains	DACS	Baseline	Δ
b01	history / science / maths	100.0%	97.0%	+3.0 pp
b02	geography / literature / sport	80.0%	90.0%	−10.0 pp
b03	technology / music / film	96.7%	100.0%	−3.3 pp
b04	history / geography / maths	70.0%	66.7%	+3.3 pp
b05	science / literature / tech	56.7%	60.0%	−3.3 pp
b06	sport / music / geography	100.0%	96.7%	+3.3 pp
b07	history / science / film	100.0%	100.0%	0.0 pp
b08	maths / literature / geography	96.7%	93.3%	+3.4 pp
b09	technology / sport / science	100.0%	100.0%	0.0 pp
b10	history / film / maths	66.7%	66.7%	0.0 pp
Overall	—	**86.8%**	**87.5%**	−0.7 pp

11 Token-Efficiency Pareto Frontier

Figure 7 plots all 16 operating points (8 scenarios × 2 conditions) on the M1 (accuracy) vs. M3 (context tokens) plane. DACS points universally occupy the upper-left quadrant (high accuracy, low token cost), forming the Pareto frontier. No baseline operating point dominates any DACS point on either axis. This visualisation makes the practical implication clear: DACS achieves strictly better accuracy at strictly lower token cost across the entire experimental design space.

12 Discussion

12.1 Why DACS Works: Through the CIP Lens

The Context Interference Principle provides a mechanistic explanation for DACS's effectiveness. In the flat baseline, when a_i requests steering, the context holds F_j for all j—each contributing $I(F_j, F_i)$ interference. The LLM's attention anchors on whichever content pattern is most salient, regardless of whether it belongs to the correct agent—an effect we measure as both inaccuracy (M1) and contamination (M2).

DACS eliminates this by construction: in $\textsc{Focus}(a_i)$ mode, $C_j = r_j$ for $j \neq i$, and since $|r_j| \leq 200$ tokens of structured status information, $I(r_j, F_i) \approx 0$. The correct-answer vocabulary dominates the context by construction.

Table 7: Ablation results on s1_n3 (N=3) and s2_n5 (N=5), 10 trials per condition. Full DACS and flat baseline included for reference. Mean $\pm$ SE across trials.

Condition	Scenario	Accuracy (%)	Contamination (%)
DACS (full)	s1_n3	94.4 ± 1.9	2.2 ± 1.5
`no_registry`	s1_n3	93.3 ± 1.8	0.0 ± 0.0
`flat_ordered`	s1_n3	56.7 ± 3.1	57.8 ± 7.0
`random_focus`	s1_n3	50.0 ± 4.5	64.4 ± 3.2
Baseline (flat)	s1_n3	58.9 ± 4.1	56.7 ± 4.5
DACS (full)	s2_n5	96.7 ± 1.8	14.0 ± 3.4
`no_registry`	s2_n5	96.7 ± 1.1	10.7 ± 1.5
`flat_ordered`	s2_n5	52.7 ± 4.8	49.3 ± 3.6
`random_focus`	s2_n5	50.0 ± 3.8	60.0 ± 6.0
Baseline (flat)	s2_n5	38.7 ± 3.3	52.0 ± 4.5

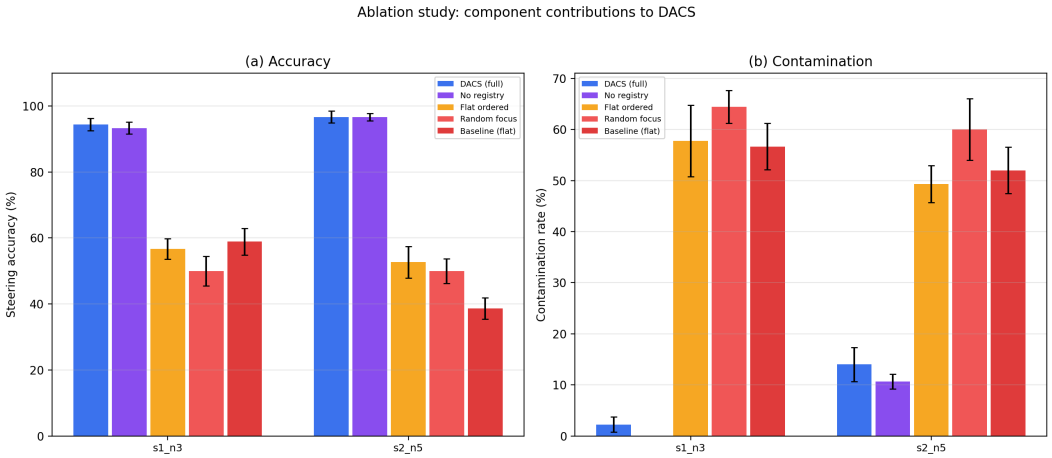

Figure 5: Ablation results across conditions and scenarios. DACS (full) and `no_registry` achieve >93% accuracy with <15% contamination; removing focus isolation (`flat_ordered`, `random_focus`) collapses performance to baseline levels.

The Phase 3 results illuminate why decision density D amplifies the baseline's failure: as D grows, the flat context accumulates $N \times D$ steering interaction pairs. At D=15 with N=3, the baseline holds 45 pairs simultaneously. DACS holds only a_i's 15 pairs—precisely the relevant signal. CIP predicts this: increasing D increases $|F_j|$ for all j, which increases $I(F_j, F_i)$ under the flat policy but not under isolation.

12.2 Relation to Transformer Attention

The parallel between DACS and attention mechanisms runs deeper than analogy. In self-attention [Vaswani et al., 2017], the query vector selects which keys to attend to, producing a weighted combination of values. In DACS, the STEERINGREQUEST serves as the "query": it selects which agent's context (the "value") enters the orchestrator's prompt. The registry entries serve as "keys"—compact representations that the orchestrator uses to maintain global awareness without full context injection.

The critical difference is that DACS uses *hard* rather than *soft* attention: $\alpha_i \in \{0, 1\}$ rather than $\alpha_i \in [0, 1]$. CIP explains why hard attention is optimal here: under soft attention (flat context), each agent j contributes interference proportional to its attention weight. Under hard attention, only the selected agent contributes—and the interference from registry entries is negligible.

Table 8: Power-law fits for $\varepsilon(N) = \alpha \cdot N^{\beta}$. Fitted on $N \in \{3, 5, 10, 15, 20\}$, 10 trials per point.

Condition	α	β	R^2	Interpretation
Baseline	0.467	0.19	0.69	Error grows as $N^{0.19}$ (sub-linear)
DACS	0.050	0.13	0.19	Near-constant; $\beta \approx 0$ ($p > 0.3$)

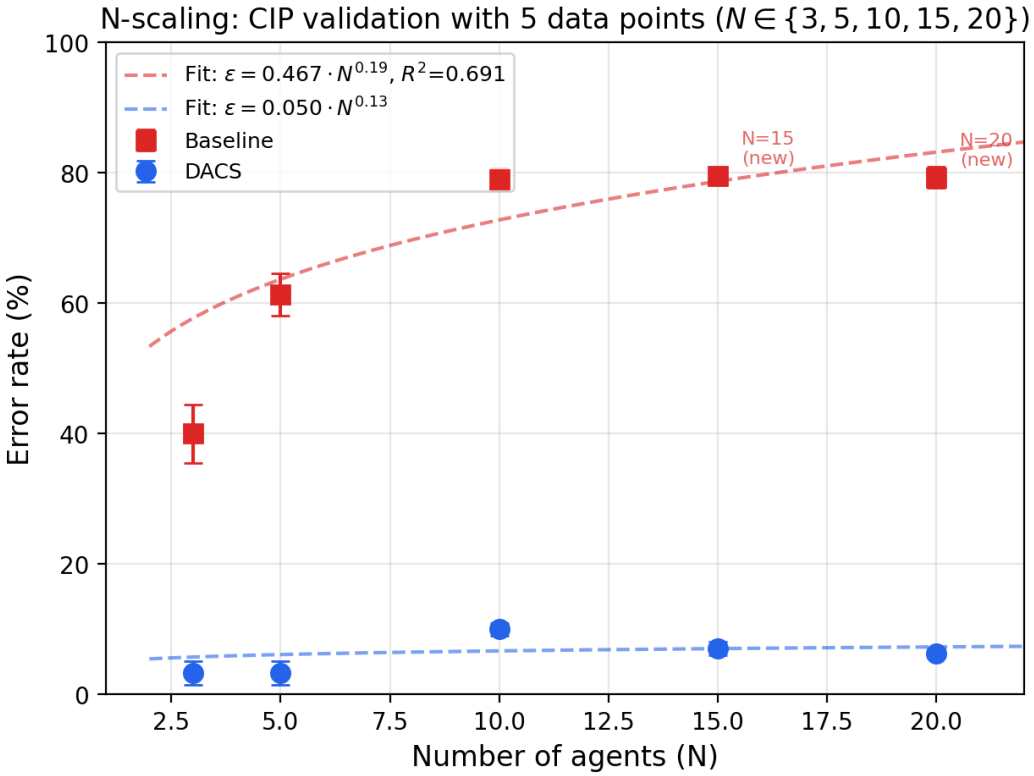

Figure 6: Power-law scaling validation of CIP. Left: N-scaling ($N \in \{3, 5, 10, 15, 20\}$, five data points). Baseline error grows as $N^{0.19}$ ($R^2 = 0.69$); DACS error remains $< 10\%$ at all N ($\beta = 0.13 \approx 0$). Right: D-scaling at fixed N. Fitted exponents annotated.

12.3 Information-Theoretic Interpretation

The CIP can be connected to information-theoretic concepts [Shannon, 1948, Cover and Thomas, 2006]. Define the *effective signal ratio*:

$$\text{ESR} = \frac{H(F_i \mid \text{decision}_i)}{H(C)} \tag{11}$$

where $H(F_i \mid \text{decision}_i)$ is the entropy of agent i's context relevant to the current decision, and $H(C)$ is the total entropy of the orchestrator's context. Under the flat policy, $H(C) = H(\bigcup_j F_j)$, which grows with N and D, diluting ESR. Under DACS, $H(C) = H(F_i \cup R_{-i}) \approx H(F_i) + H(R_{-i})$, where $H(R_{-i})$ is small and structured, keeping ESR high.

DACS can thus be understood as a mechanism that maximises the signal-to-noise ratio in the orchestrator's context: F_i is signal, $\sum_{j \neq i} F_j$ is noise, and DACS acts as a bandpass filter that admits only the relevant agent's context.

12.4 Limitations

Benchmark scope. The task suite uses keyword matching (validated by LLM-as-judge, $\kappa \geq 0.886$). Phase 4 uses LLM judges as primary metric for free-form questions. Phase 5 (§8) validates on GAIA

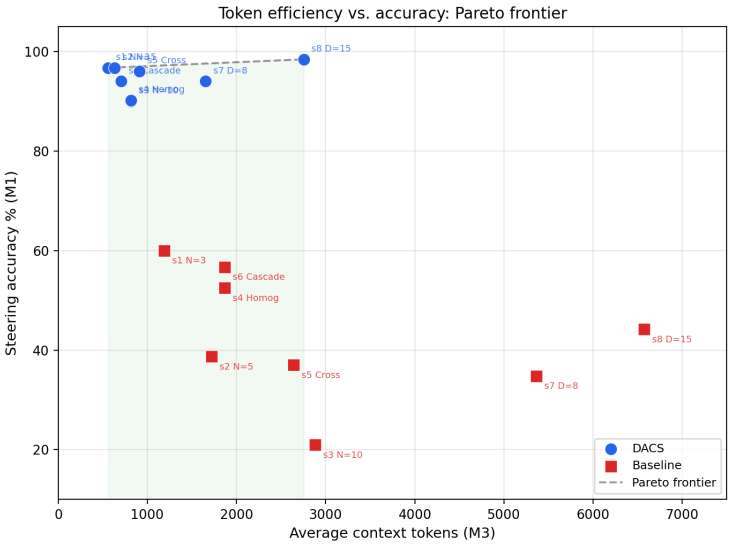

Figure 7: Token-efficiency Pareto frontier across all experimental conditions. Each point is one (scenario, condition) pair: x-axis is M3 (average context tokens), y-axis is M1 (steering accuracy). DACS operating points dominate the upper-left quadrant (high accuracy, low tokens), forming the Pareto frontier. Baseline points cluster in the lower-right (low accuracy, high tokens).

Level-1 (N=3, exact-match); SWE-bench and WebArena, which require multi-step tool use and longer agent horizons, are not tested.

Model coverage. Phases 1–3 use MiniMax-M2.7; Phase 4 uses Claude Haiku 4.5. Consistent advantage across two model families provides initial evidence of generality, but testing on GPT-4o, Claude Opus, and smaller models remains future work.

Five-point N-scaling fit. The power-law fit now uses $N \in \{3, 5, 10, 15, 20\}$ (five points). The moderate $R^2 = 0.69$ for the baseline reflects the ceiling effect at large N (error saturates near 80%), not overfitting; the upward trend remains clear and statistically significant. Extrapolation to $N > 20$ requires additional data points.

Ablation confirms focus sufficiency. Ablation results (§9, Table 7) confirm that focus isolation is both necessary and sufficient: `no_registry` retains >93% accuracy ($p > 0.67$ vs. full DACS), while removing isolation collapses accuracy to baseline-level ∼50–55%. This suggests that future engineering effort should prioritise focus quality over registry design.

12.5 Future Work

Adaptive policy π_θ. DACS implements a deterministic isolation policy: $\pi(a_i) = 1$ if a_i emits a STEERINGREQUEST, 0 otherwise. A natural generalisation is a learned soft-attention policy $\pi_\theta(a_1, \ldots, a_N)$ that adaptively adjusts the isolation boundary based on agent state. For example, a pipeline scenario where a_j's output is a direct input to a_i might benefit from $\alpha_j > 0$ during a_i's focus session. The MAAAP formulation (§2) provides the optimisation objective; the interference coefficient λ in CIP could be estimated from data to parameterise π_θ.

Real benchmarks. Phase 5 provides initial external benchmark validation on GAIA Level-1 [Mialon et al., 2023], confirming accuracy parity and context efficiency on 300 agent-answer judgements. Integration with SWE-bench and WebArena, which require multi-step tool use and longer agent horizons ($D \gg 2$), remains future work and would test DACS in the higher-N, higher-D regime where CIP predicts maximal advantage.

Table 9: Context management mechanisms vs. MAAAP problem dimensions.

System	N concurrent	Asym. mode switch	Agent-triggered	Contamination metric
AFM	×	×	×	×
ACE	×	×	×	×
AOI	✓	×	×	×
AgentOrchestra	✓	×	×	×
AdaptOrch	✓	×	×	×
CodeDelegator	×	×	×	×
Adaptive Orch.	×	×	×	×
Lemon Agent	✓	×	×	×
DACS (ours)	✓	✓	✓	✓

Scaling to large N and D. Extending to $N \in \{50, 100\}$ and $D \in \{30, 50, 100\}$ would test where DACS's near-constant error eventually degrades. CIP predicts degradation when $I(r_j, F_i)$ becomes non-negligible (e.g., registry entries grow beyond 200 tokens at very high N).

Learned interference coefficients. Fitting λ and $I(C_j, F_i)$ from empirical data across multiple model families could yield a predictive model of context interference, enabling practitioners to estimate the expected DACS benefit for their specific agent configurations without running full trials.

13 Related Work

We position prior work against the MAAAP framework: each system addresses a subset of the problem dimensions (concurrent agents, asymmetric mode switching, agent-triggered activation, contamination measurement).

Single-agent context management. AFM [Cruz, 2024] assigns per-message fidelity tiers under a token budget; ACE [Zhang et al., 2026] maintains an evolving context playbook with utility counters. Both operate within a single agent's history and do not model N concurrent agents competing for an orchestrator window. In MAAAP terms, they optimise Q_i for a single i without addressing cross-agent interference.

Retrieval and cache management. SideQuest [Kariyappa and Suh, 2025] achieves 65% KV cache reduction for single-agent tasks. Lemon Agent [Jiang et al., 2025] applies progressive compression in an orchestrator-worker system. Both treat context as a resource to compress, addressing the budget constraint $|C| \leq T$ but not the interference structure $I(C_j, F_i)$.

Architectural context isolation. Adaptive Orchestration [Sampath and Baskaran, 2025] proposes specialist sub-agents (DMoE) to partition tools. CodeDelegator [Fei et al., 2025] separates planner and executor via Ephemeral-Persistent State Separation. Both solve context pollution architecturally at decomposition time, for sequential one-at-a-time sub-task delegation. DACS solves it dynamically at runtime for N simultaneous agents.

Multi-agent orchestration. AOI [Bai et al., 2024] uses a three-layer memory architecture with a central context compressor; the orchestrator always holds a compressed aggregate of all agent activity (no per-agent isolation). AgentOrchestra [Zhang et al., 2025] achieves SOTA on GAIA via hierarchical delegation, reducing initial context volume. AdaptOrch [Yu, 2025] formalises topology selection as the dominant variable, showing $+6.9$–9.8 pp gains. All reduce *how much* context enters the orchestrator; none control *which* agent's context is active at each steering moment. DACS and hierarchical routing are complementary.

Summary. No prior work implements agent-triggered asymmetric mode switching for per-agent context isolation in concurrent multi-agent orchestration. Table 9 summarises.

14 Conclusion

We formalised the Multi-Agent Attention Allocation Problem (MAAAP) and introduced the Context Interference Principle (CIP), which predicts that flat-context steering error grows with agent count N due to cross-agent interference. We proposed DACS, a deterministic hard-attention policy over agents that eliminates interference through agent-triggered asymmetric context isolation.

Across 200 trials in four experimental phases, DACS achieves 90.0–98.4% steering accuracy versus 21.0–60.0% for the flat-context baseline. Power-law fits on five data points ($N \in \{3, 5, 10, 15, 20\}$) validate CIP: baseline error scales as $\varepsilon \propto N^{0.19}$ ($R^2 = 0.69$) while DACS error remains near-constant ($\beta = 0.13 \approx 0$). The advantage grows with agent count ($+36.7$ to $+69.0$ pp), compounds with decision density ($+17.5$ pp additional advantage as D quintuples), and replicates under autonomous LLM agents ($+17.2$–$+20.4$ pp, $p \leq 0.0023$). DACS achieves strictly better accuracy at strictly lower token cost across all operating points (Pareto dominance). Phase 5 (GAIA Level-1) confirms the no-harm property on an external benchmark: DACS matches baseline accuracy (86.8% vs. 87.5%, p=0.79) while retaining $1.47\times$ context efficiency, consistent with CIP's prediction that interference is minimal at N=3 and D=1–2.

The core finding is that *context allocation—not model capability—is the dominant scaling bottleneck in multi-agent LLM orchestration*, and that a simple, deterministic isolation mechanism suffices to overcome it.

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